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THE HISTORY OF

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PHARO

A clean-room fork of Squeak aimed at professional software engineering, research, and industrial use

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WHY FORK SQUEAK?

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By the late 2000s Squeak was simultaneously a children's medium, a research playground, and the only serious free Smalltalk -- and those constituencies pulled in different directions. In 2008 Stephane Ducasse and Marcus Denker announced Pharo [1]: a fork whose goal was not media or pedagogy but a clean, MIT-licensed, professional Smalltalk platform. The early releases were largely subtractive: Etoys and accumulated experiments were removed, the licensing of every remaining method was audited, and the kernel was shrunk toward something bootstrappable and explainable.

The name signals the intent -- a pharos, a lighthouse: an environment meant to guide serious engineering. Where Squeak's loyalty was to the Dynabook, Pharo's was to the working programmer, while keeping the property that makes Smalltalk Smalltalk: a fully live system of objects you inspect, modify, and debug while it runs.

WHAT PHARO CHANGED

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Pharo rebuilt the tower floor by floor while living in it. The compiler, the class builder, and reflection were reworked (slots, first-class instance variables, metalinks); the Morphic UI gave way to new graphics stacks (Bloc) and the Glamorous Toolkit school of "moldable" tools grew out of Pharo's inspector. Iceberg brought native Git integration -- ending the era when Smalltalk code lived only inside images -- and the image itself became reproducible: modern Pharo is bootstrapped from source, not descended by copy from the 1970s PARC image.

The open books mattered as much as the code: Pharo by Example and its successors, free from the Pharo books collection [2], gave the platform the approachable canonical documentation commercial Smalltalks never had. A consortium of companies and universities [3] funds engineers, which has kept releases on an annual cadence (Pharo 12 and beyond) across Intel and ARM, desktop and headless server.

RESEARCH VEHICLE AND INDUSTRIAL TOOL

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Pharo became the standard vehicle for a strand of European programming-language and software-analysis research: Moose (software analysis), Roassal (visualization), and hundreds of papers use it as their substrate, continuing the Smalltalk tradition of the system as laboratory. On the industrial side it powers domain tools in finance, logistics, and modeling -- often through Seaside and Teapot web stacks or through GemStone/S compatibility for object databases.

In the arc of this series Pharo is the demonstration that the 1970s live-object medium is not a museum piece: stripped, relicensed, re-bootstrapped, and pointed at Git and CI, the same fundamental design -- everything is an object, the tools are

made of the same stuff as the programs, the system is always running -- competes on modern terms. The lighthouse is the old PARC light, re-housed.

SOURCES & CITATIONS

- [1] Pharo official site: the immersive programming experience
<https://pharo.org/>
- [2] Pharo open books collection (Pharo by Example, Pharo with Style, and others), free downloads
<https://books.pharo.org/>
- [3] Pharo Consortium: the funding and governance body
<https://consortium.pharo.org/>
- [4] Pharo source repository
<https://github.com/pharo-project/pharo>
- [5] Stephane Ducasse's free Smalltalk books archive (including the Smalltalk-80 Blue Book lineage Pharo descends from)
<http://stephane.ducasse.free.fr/FreeBooks.html>
- [6] Moose: software and data analysis platform built on Pharo
<https://moosetechnology.org/>

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